History

In 1948, the Air Force (AF) commissioned the Rand Corporation in Santa Monica, California to recommend development work in reconnaissance satellite programs known as Project Feedback.
In 1949, the Rand Corporation produced a report the reached two key conclusions:

- A radioactive cell–mercury vapor system was technically feasible for space power
- Such a system could supply 500 watts of electric power for up to one year aboard a satellite

In 1951, the Atomic Energy Commission (AEC) commissioned formal industry studies for a 1-kilowatt electrical space power plant using either reactors or radioisotopes. Several companies concluded isotopes were the preferred approach.
In 1952, RAND issued a report discussing radioisotopic power for space power.
In 1954, the Pied Piper Program conducted specialized studies on nuclear-electric sources for space application.
With reconnaissance satellite power needs now clearly defined, the Department of Defense formally requested in August 1955 that the AEC develop a nuclear reactor auxiliary power unit for the Air Force satellite program. The AEC agreed and stated it would "explore the possibilities of using both radioisotopes and reactors as heat sources". The program ran under the classified code name "Pied Piper" between 1955 and 1957. It was later changed to SNAP (Systems for Nuclear Auxiliary Power).
The AEC began parallel power plant efforts with two private corporations:

- Odd-numbered SNAP programs using radioisotopes were spearheaded by contractual work at the Martin Company.
    - SNAP-1: This was an early effort by the Martin Company to use the decay heat of cerium-144. The design involved using this heat to boil liquid mercury to drive a small turbine. It was eventually superseded by static conversion technologies that did not require moving parts.
    - SNAP-3 / SNAP-3A: Known as the "salesman" of the program, SNAP-3 was a proof-of-principle device utilizing a thermopile to convert heat directly into electricity.
        - Public Debut: President Eisenhower unveiled the 5-pound "atomic battery" in the Oval Office on January 16, 1959.
        - First Use in Space: A modified version, SNAP-3A, fueled by plutonium-238, was launched aboard the Navy Transit 4A satellite on June 29, 1961. This marked the first time a nuclear power source was used in space, providing roughly 2.7 watts of electrical power.
    - SNAP-7: This series consisted of prototype units fueled with strontium-90. Unlike the space-focused units, these were designed for terrestrial applications in severe environments, such as powering Navy and Coast Guard navigation aids and automatic weather stations.

    - SNAP-9A: Designed for operational Navy Transit satellites, these were significant for providing the entire electrical power for a spacecraft.
        - Power: These units provided approximately 25 watts and were designed for a 5- to 10-year lifespan.
        - History: Two units successfully achieved orbit in 1963. However, a mission abort in April 1964 resulted in a unit burning up and dispersing in the upper atmosphere as designed, which later prompted a shift toward "intact re-entry" safety concepts for future units.
- Even-numbered SNAP reactor power systems were developed through contractual work with the Atomics International Division of North American Aviation, Inc.
    - SNAP-2: This was a Zirconium-Hydride (Zr-H) reactor system designed to produce 3 to 5 kWe.
        - Technology: It utilized a Mercury-Rankine power conversion cycle where sodium-potassium (NaK) coolant transferred heat to mercury vapor to drive a Combined Rotating Unit (CRU).
        - Testing: Two prototypes were tested: the SNAP Experimental Reactor (SER), which operated from 1959 to 1961, and the SNAP 2 Developmental Reactor (S2DR), which operated from 1961 to 1962. The program was cancelled in 1963 due to budget cuts.
    - SNAP-8: A larger-scale joint effort between the AEC and NASA, designed to produce 35 kWe (later pushed toward 50 kWe).
        - Design: Like SNAP-2, it used a Zr-H reactor and a Mercury-Rankine cycle, but with a much higher performance goal of 600 kWt thermal power.
        - Prototypes: Testing was conducted on the SNAP 8 Experimental Reactor (S8ER) and the SNAP 8 Developmental Reactor (S8DR). Both tests concluded with the discovery of cracked fuel cladding, highlighting technical hurdles in the high-power reactor program.
    - SNAP-10A: This remains the only nuclear reactor the United States has ever flight-tested.
        - Technology: It was a 500-watt Zr-H reactor that utilized thermoelectric conversion instead of a rotating turbine.
        - Flight Mission: Launched on April 3, 1965, it achieved a circular polar orbit and operated successfully for 43 days.
        - Shutdown: It was shut down prematurely due to a high-voltage failure in the spacecraft's electrical system (the voltage regulator), rather than a failure of the reactor itself. Ground testing of a twin unit continued for 10,000 hours without interruption.

SNAP 10A

Overview

The Systems for Nuclear Auxiliary Power (SNAP) 10A was a compact nuclear reactor system developed to provide reliable, long-term electrical power for space applications. Designed as a Zirconium-Hydride (Zr-H) reactor, its primary objective was to produce a minimum of 500 watts of electrical power (We) for an operational lifespan of one year or longer.

SNAP 10A holds the distinction of being the only nuclear reactor the United States has ever flight-tested. It was successfully launched on April 3, 1965, and placed into a circular polar orbit. The system operated successfully for 43 days before a high-voltage failure in the electrical system of the Agena spacecraft, not the reactor itself, caused a premature shutdown. Despite this, the mission was considered a success, demonstrating that a reactor could be safely launched, automatically brought to power in orbit, and operated as predicted by ground testing.

Full element design

Two major environments were considered in designing the SNAP-10A reactor: launch and orbital operation.

Basic design requirements:

1. Productions of 500 W at 28 vd-c for one year in space.
2. Capability of remote startup in orbit by ground command.
3. Capability for handling, transport, and launch within acceptable safety criteria and currently accepted launch base procedures.
4. No requirement for long-term dynamic control.
5. Sufficient instrument capability on the SNAP-SHOT launch to monitor system performance fully.
6. Radiation shielding to attenuate realtor leakage to acceptable levels within the spacecraft.

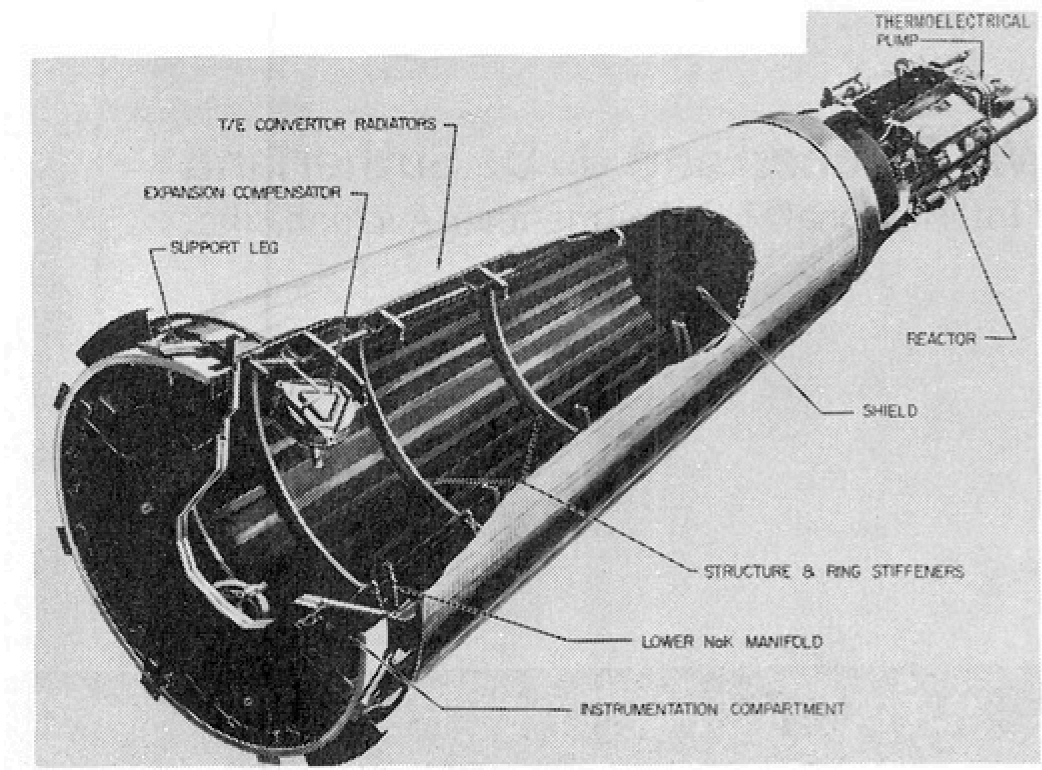


Fig. 1 - SNAP-10A system

The reactor is the ZrH type with about 4.8kg of fully enriched 235-Uranium fuel mixed into the metal hydride.

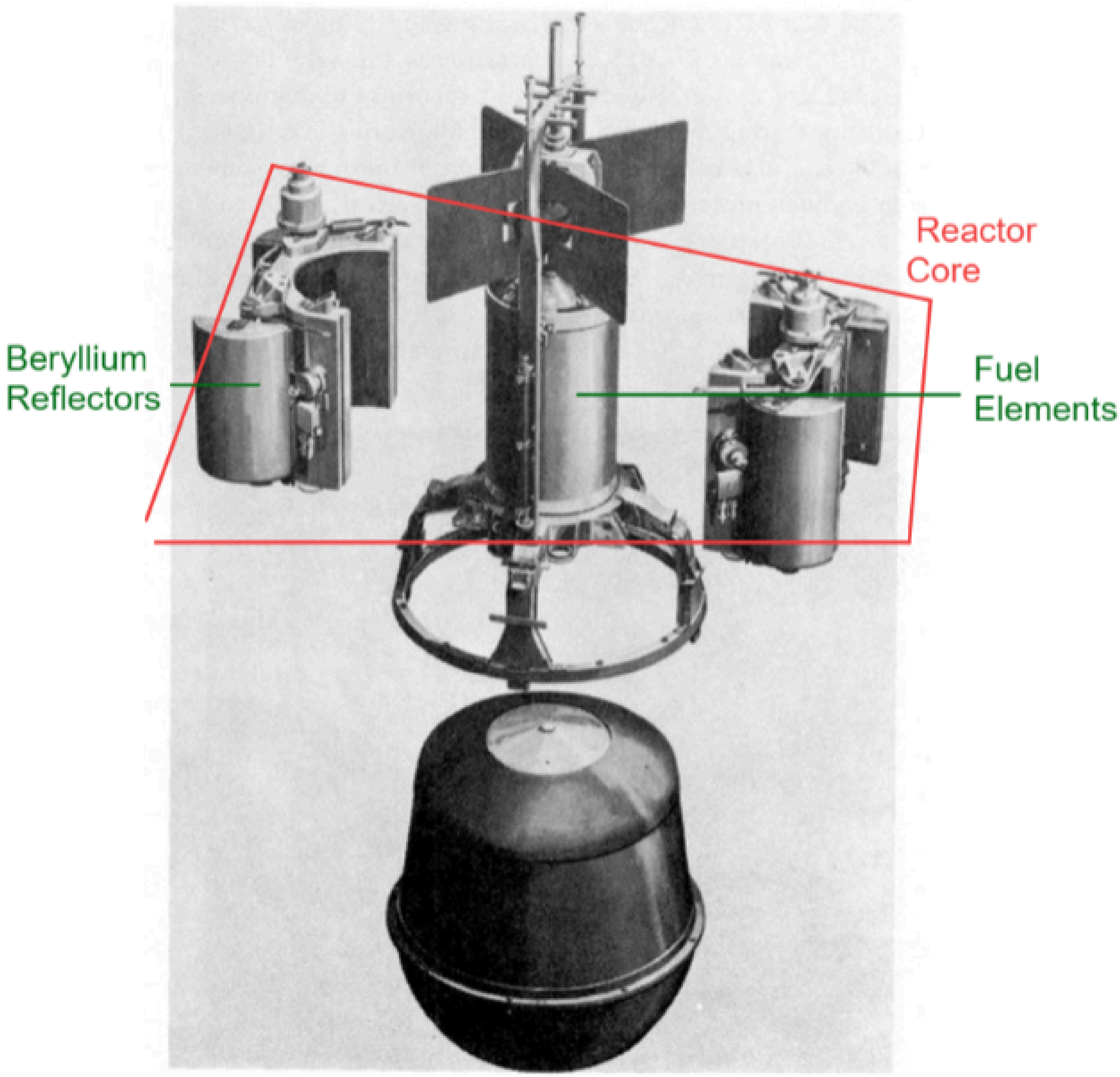
The SNAP 10A consists of 5 major components:

1. The reactor structure
2. Radiation shield
3. Reflector assembly
4. Reactor control equipment
5. Reactor core

- 37 fuel elements, each 32.64 cm and 3.17 in diameter.

<u>SNAP-10A Reactor Core</u>

The SNAP-10A core is a right circular cylinder consisting of a hexagonal array of 27 fuel elements surrounded by Beryllium reflectors to form the cylindrical geometry.



Each fuel element is a cylinder 12.450 in. long with an outside diameter of 1.25 in.
The fuel material is a a zirconium-10 weight % uranium alloy hydrided to a nominal N_H of 6.35 (that is, 6.35×10^{22} atoms of hydrogen per cc of fuel). Each element contains approximately 128 grams of U^{235}, 10 grams of U^{238}, 25 grams of hydrogen, and 1240 grams of zirconium.

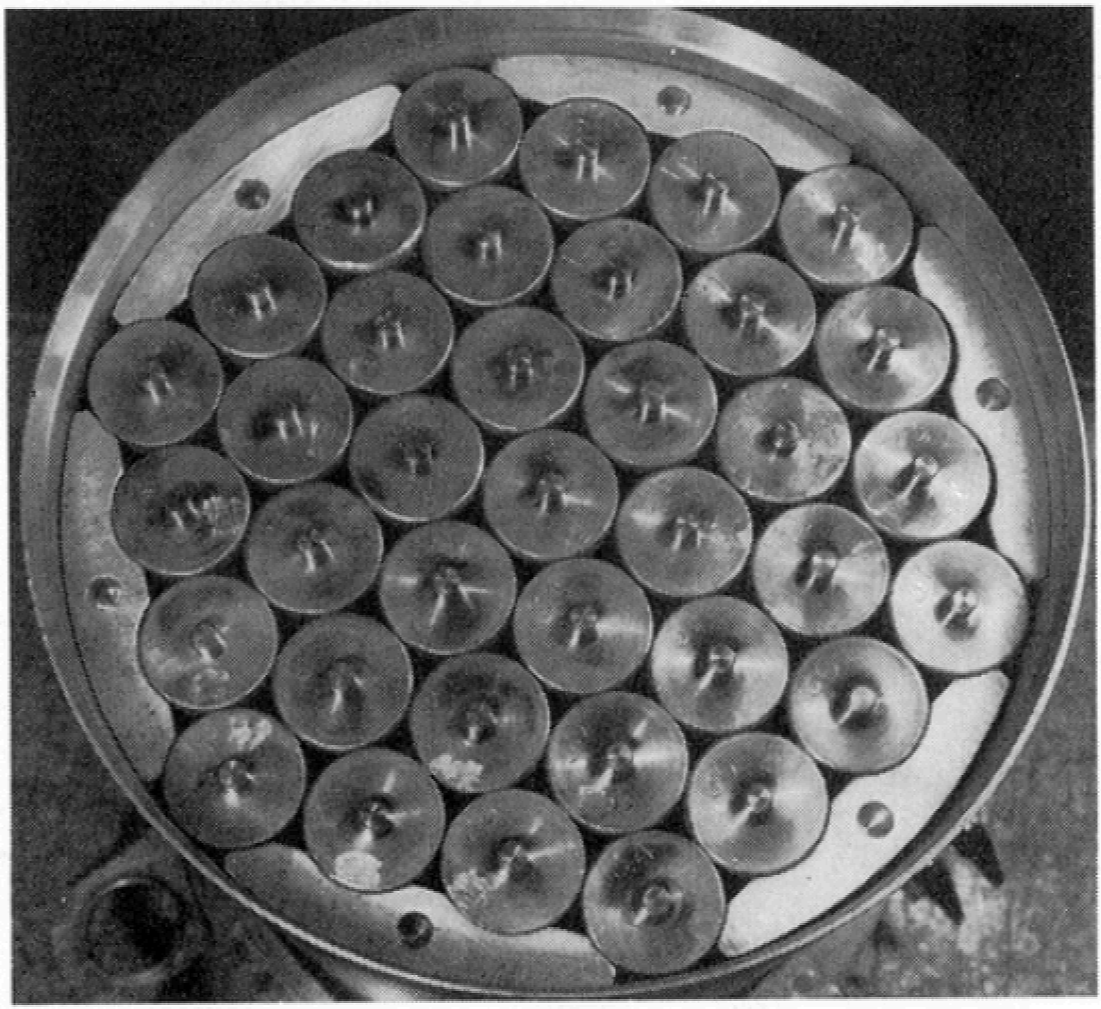


Fig. 6 - SNAP-10A reactor core (top view)

The fuel elements are clad in Hastelloy-N tubes having a wall thickness of 0.015 in. Hastelloy-N was selected as the cladding material because of its superior creep properties at operating temperatures to the 300 series stainless steels, and also because its coefficient of thermal expansion is much closer to that of the fuel material.

- **Cladding material** refers to the outer layer of material applied over another surface to serve as a protective, insulating, and decorative skin

The ends of the fuel elements are sealed with Hastelloy-N caps welded to the cladding tube. End pins, which position the fuel elements in the grid plates, are integral with the end caps. The pins on each end are of different diameters, ensuring proper orientation of the elements. Internal surfaces of the cladding tube and end caps are coated with a 0.003 in. layer of a ceramic material. This coating acts as a barrier, preventing hydrogen leakage from the fuel material in the high-temperature and high-radiation environment of the reactor core. The weight of each fuel element is 3.4 lb.

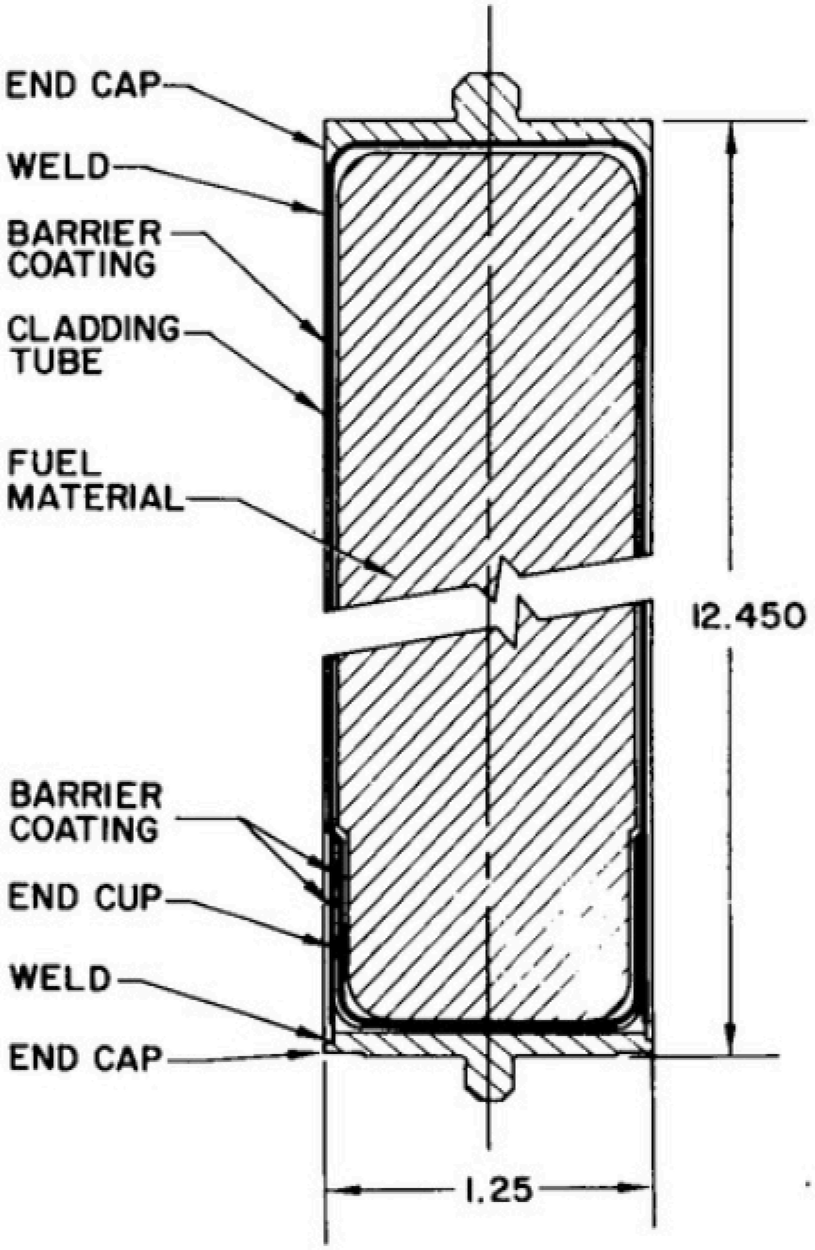


Fig. 7 - SNAP-10A fuel element

Six internal neutron reflectors are six solid blocks of Beryllium. Holes in each end of the reflectors are used for positioning. The reflectors have a nominal length of 12,455 in. The reflectors carry the precompression load of the core hold-down springs. Each reflector weighs 1.5 lb.

The upper and lower grid plates support and position the fuel elements and internal reflectors within the reactor vessel. Both plates are circular, 8.750 in. in diameter.

- The upper grid plate is 0.125 in. thick, made from a solid piece of type 316 stainless steel.

- The lower grid plate is a brazed assembly consisting of two 0.060 in. thick sheets with spacers between them. Overall thickness is 0.50 in. The lower plate is also type 316 stainless steel brazed with Nicrobraze 50 alloy. A composite structure was selected for the lower grid plate because it provides lighter weight than a solid plate of the same rigidity. The lower plate carries three times the load of the upper plate during the launch phase.

Each grid plate is perforated with 109 holes on a 0.728 in. triangular pitch. Thirty-seven of these holes are used for positioning the fuel elements. The remaining 72 are coolant flow holes.
Coolant flow through the core is modified by the lower grid plate. The coolant holes in the lower sheet of the lower grid plate vary from 0.250 in. (surrounding the central fuel element) to 0.188 in. (at the outermost channels). In this fashion, the coolant flow profile across the core is matched with the radial power distribution, permitting all fuel elements to operate at approximately the same temperature. The coolant holes in the upper grid plate and the upper sheet of the lower grid plate are perforated with uniform holes 0.375 in. in diameter.
Six pins are attached to each plate to position the internal reflectors. Twelve additional pins are attached to the upper grid plate to locate the core hold-down springs.

Reactor vessel

The reactor vessel is a thin-walled cylindrical shell, closed at one end. The inside diameter was determined by the core size. The overall height was established by core size plus the required NaK inlet and outlet plenum size as determined by hydraulic flow tests.
Type 316 stainless steel was chosen to be the vessel material.
The vessel shell is 0.032 in. thick in the core region, 0.063 in. thick in the region of the lower head, and 0.125 in. thick elsewhere.
A shell maximum thickness of 0.125 in. is required in the region of the top head attachment to transmit bending moments induced by external reflector and NaK pump side loads. The shell is also 0.125 in.
The reractor core is supported by the grid plate support ring. THe ring is attached to the vessel inner wall by 11 spot welds. A step is then milled in the ring forming the platform for the lower grid plate.

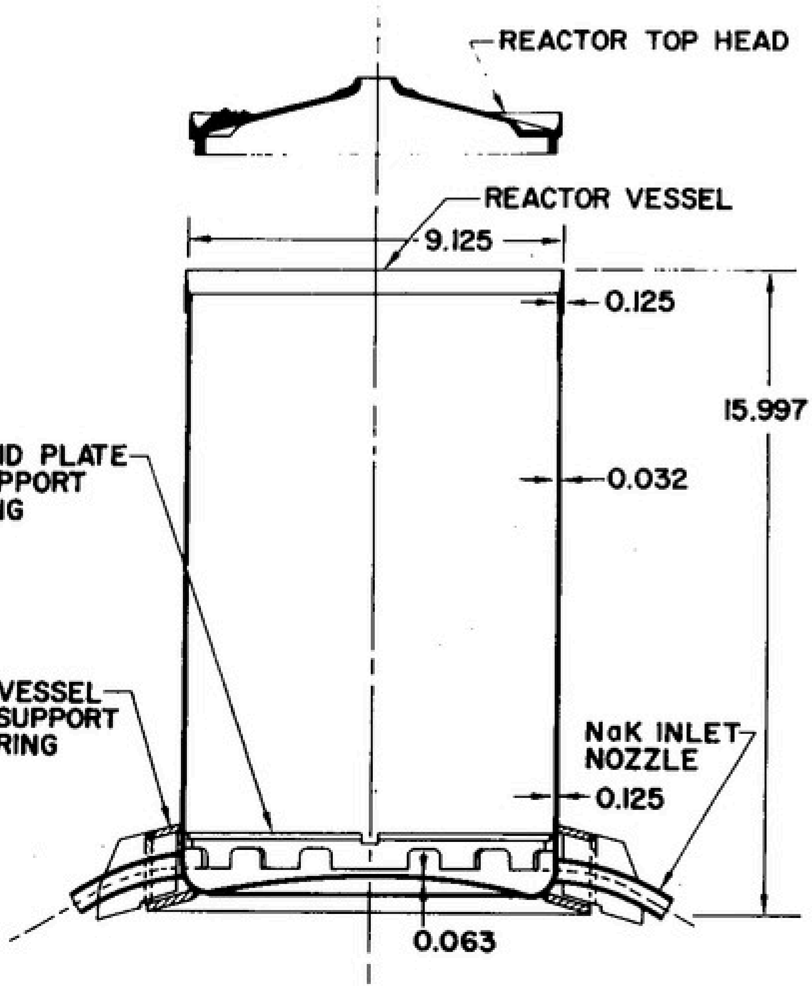


Fig. 12 - Reactor vessel assembly and top head

Two brackets, located diametrically opposite each other, are welded to the vessel in the region of the top head. These brackets serve to locate the external reflectors and distribute reflector side loads during launch. A secondary function of the stops is to provide lateral support for the NaK return tubes.

NaK inlet nozzles are welded on diametrically opposite sides of the vessel lower plenum. The nozzles are type 316 stainless steel tubing flattened to form an ob-round cross-section. Weight of the reactor vessel assembly is 26 lb.

<u>Core hold-down springs</u>

Twelve compression springs are positioned between the upper grid plate and the top head .The springs are made of Rene 41 material. The spring rate is 357 lb/in. and each spring nominally exerts a force of 35 lb on the upper grid plate. The choice of material was made because of its low relaxation rate at operating temperatures.

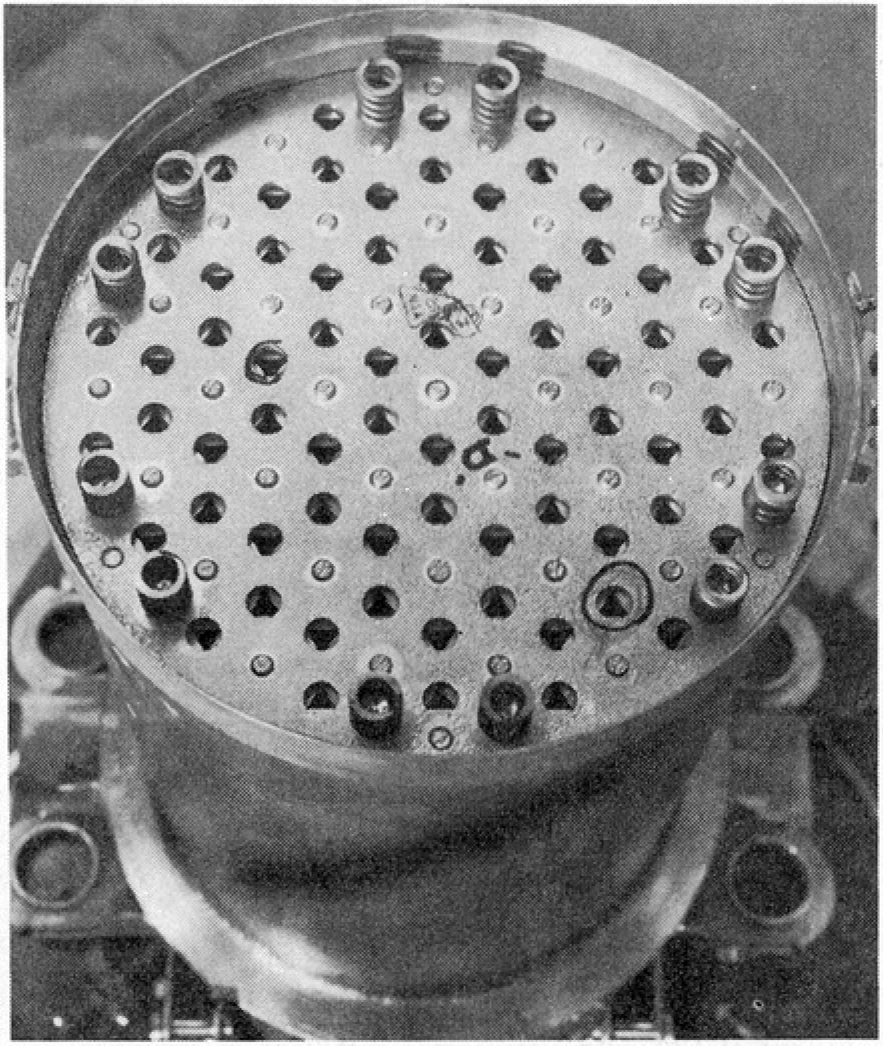
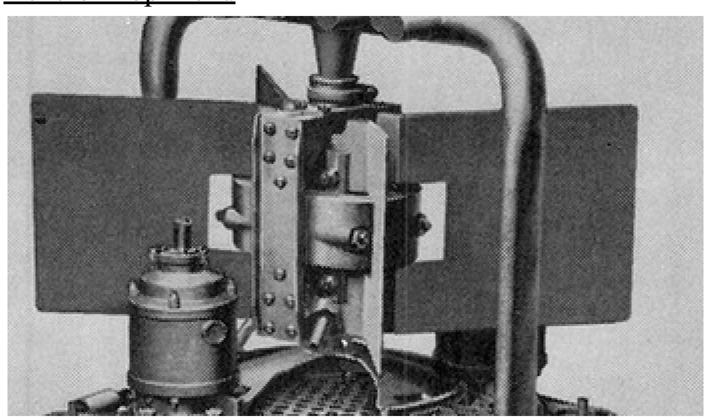


Fig. 13 - SNAP-10A reactor with top head removed

<u>Reactor Top Head</u>



The vessel top head is designed to support the thermoelectrical NaK pump and to distribute external reflector side loads. The pump support requirement necessitates a head thickness greater than that necessary for pressure containment and meteoroid protection. The head is made from a type 316 stainless steel forging.

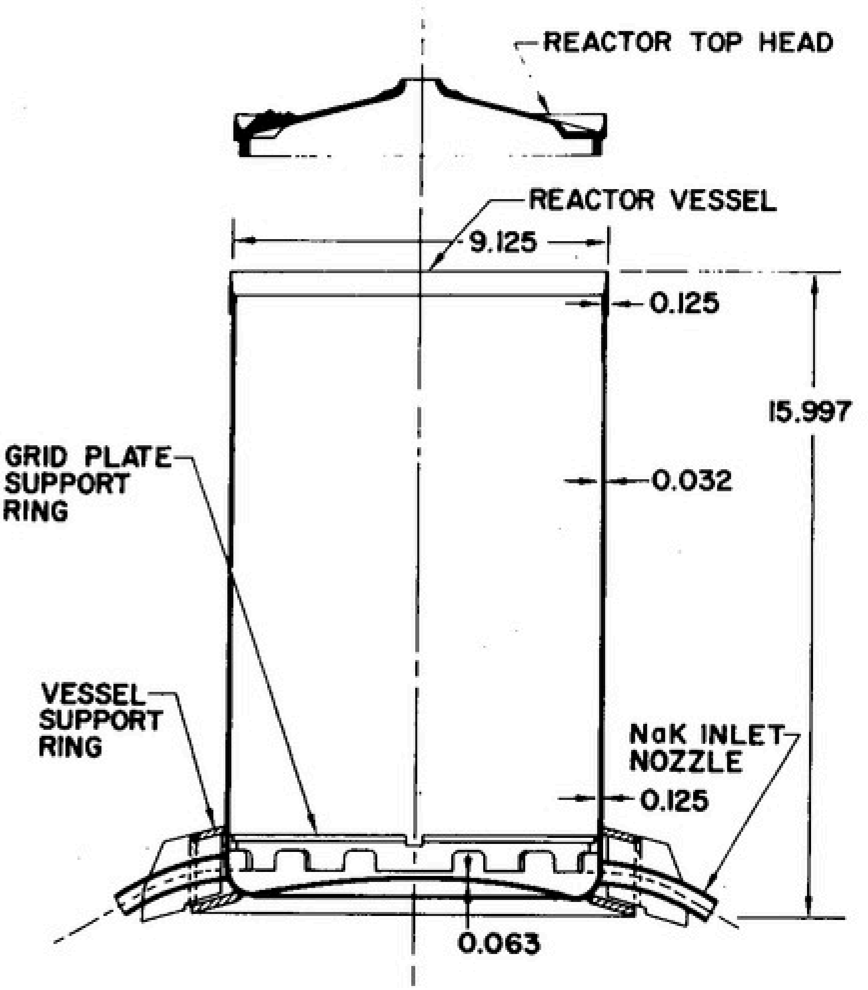


Fig. 12 - Reactor vessel assembly and top head

Reactor Support Legs

The reactor is supported from the converter structure by four support legs (Fig. 14). The support legs are designed to withstand reactor and reflector loads during the launching operation, and to permit relative thermal expansion between the reactor and converter structure during high-temperature operation.

The support legs are essentially box sections, hot formed and welded from 0.032 in., 6A1-4V titanium alloy. The support legs are fastened to the reactor vessel box ring by “blind-bolts.” Ten “blind-bolts” are used on each leg. At the lower end, the support legs are riveted to a torque box which forms part of the converter structure.

NaK inlet and outlet pipes are routed through the support legs. The legs thus afford meteoroid protection to the piping, permitting thinner walls to be used.

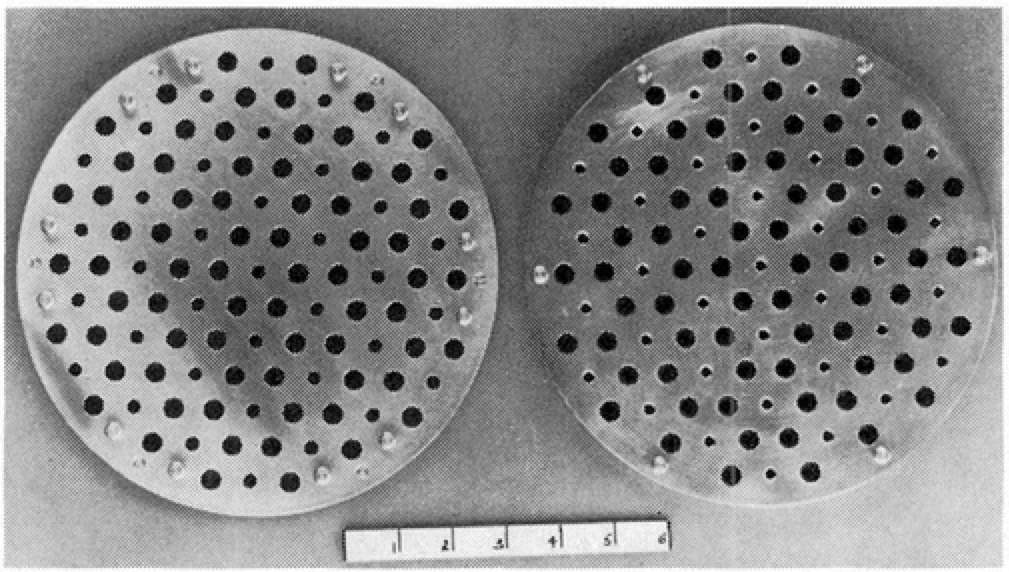


Fig. 9 - SNAP-10A grid plates

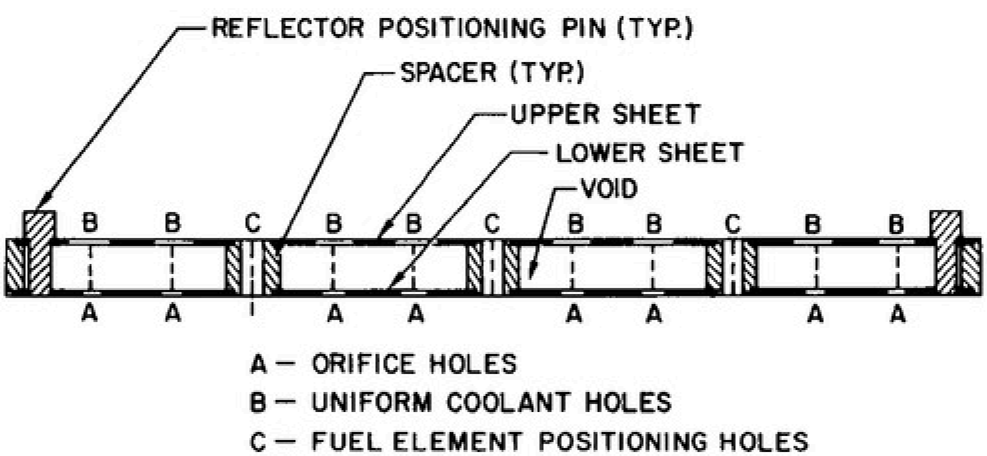


Fig. 10 - SNAP-10A lower grid plate cross-section

Reactor Reflector and Control Assembly
Heat Transfer and Power Conversion System
Shield
SNAP-10A Test History

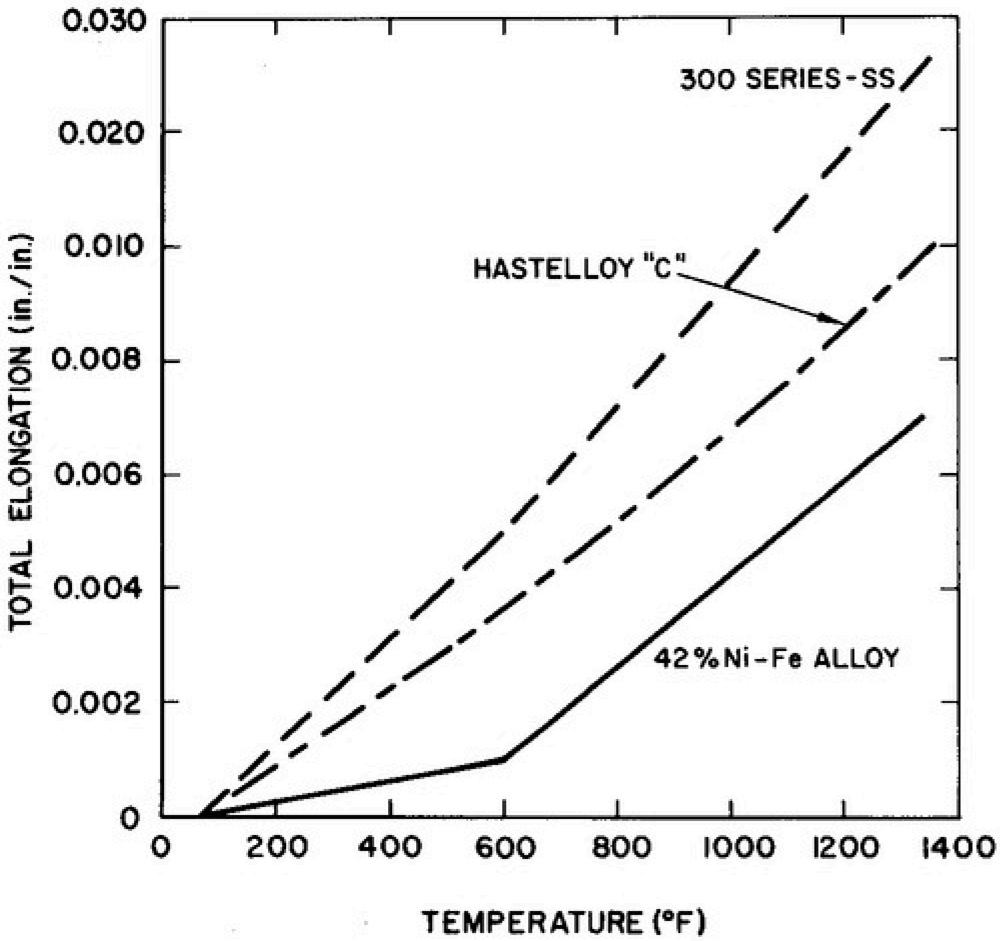


Fig. 11 - Total elongation versus temperature for possible grid plate materials

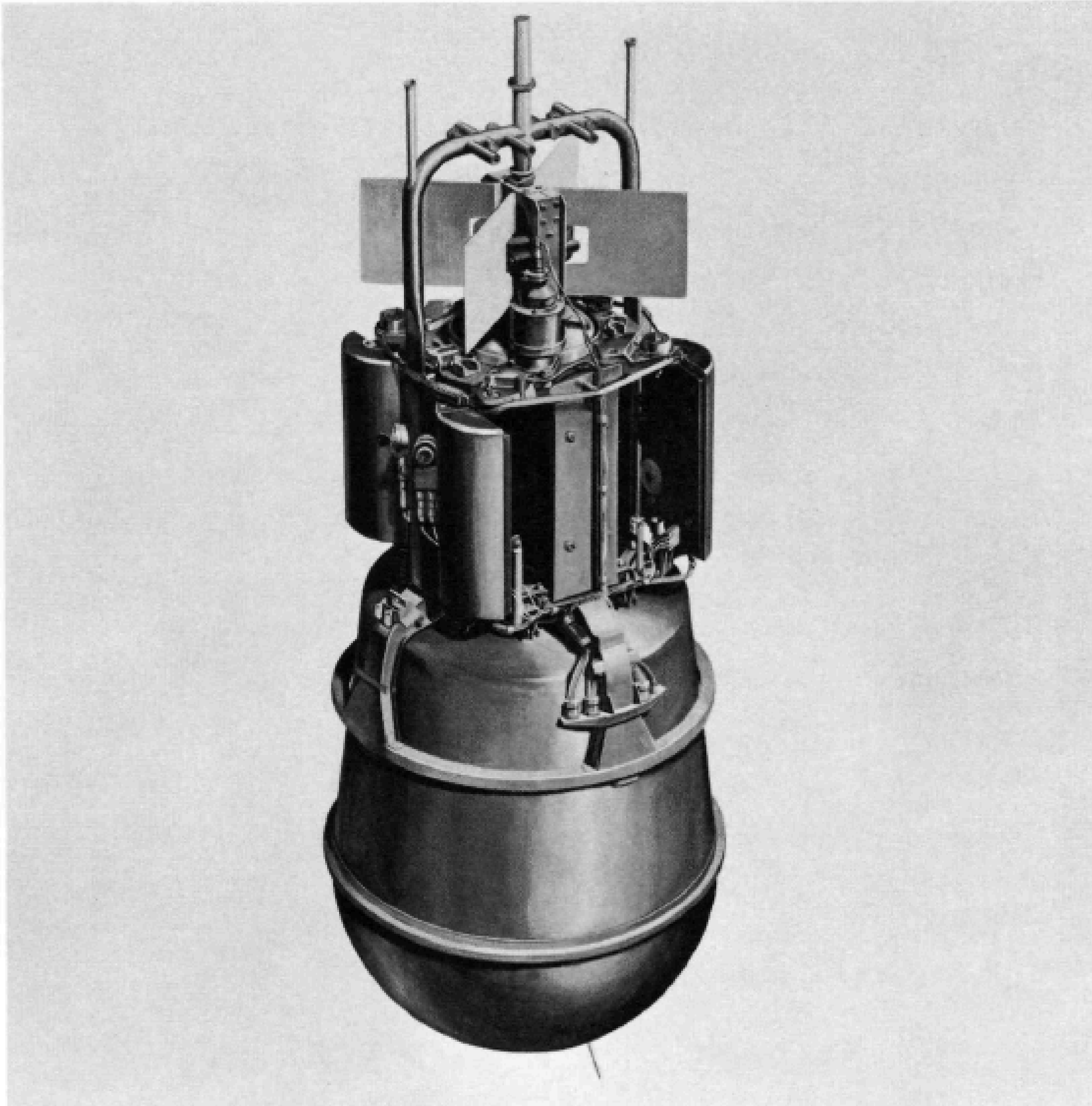


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Figure 1. SNAP 10A Reactor and Shield

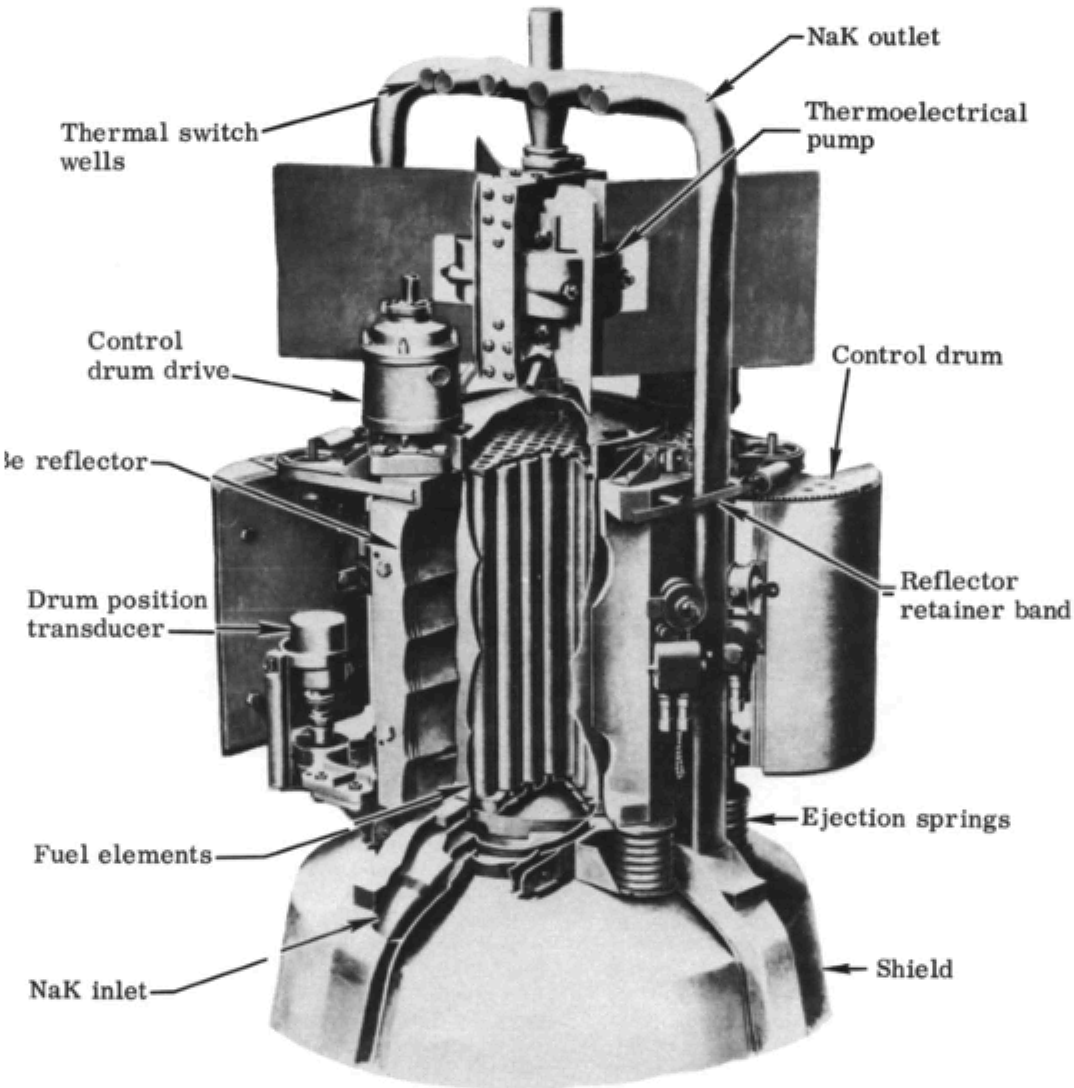

Fig. 3. SNAP 10A reactor.

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Literature Review of U.S. Space Nuclear Launch Regulation: From SNAPSHOT to NSPM-20

Citations follow the Chicago author-date system, with a reference list at the end.

Introduction and scope

The literature on how the United States authorizes the launch of nuclear systems into space falls into two bodies that rarely speak to each other. The first is the engineering and safety documentation produced around the SNAPSHOT program in the early to mid 1960s, when the SNAP-10A reactor became the only fission reactor the country has ever flown. The second is the contemporary policy and implementation literature that has grown up around National Security Presidential Memorandum 20, issued in 2019. This review covers both, treats them as a single regulatory lineage rather than two unrelated topics, and argues that the gap between them is itself the most important finding. The older material is dense with accident analysis but predates any codified approval regime, while the modern material is rich in process description but thin in independent critique, and almost nobody connects the two. The SNAP-10A precedent, which ought to be the natural test case for current rules, is invoked far more often than it is actually analyzed.

The early regime and its documentary record

The primary record of the SNAPSHOT era is largely contractor documentation produced by Atomics International for the Atomic Energy Commission, and it is surprisingly candid. The 1961 flight safety program plan lays out the safety philosophy that would govern the flight, namely launch the reactor cold, defer criticality until a long-lived orbit is confirmed, and rely on that orbit for disposal, and it is unusually honest about what had not yet been solved, especially accidental criticality on water immersion and the absence of an inherently subcritical core (Atomics International 1961). The capstone of this record is the final safeguards report, which structured its analysis in three phases running from launch through orbit injection to reactor startup and reentry, and which was submitted to the AEC, the Air Force, and the Navy Pacific Missile Range for approval (Atomics International 1965). The flight itself is documented in the open literature by the Atomics International team, whose account doubles as the post-flight report and records the voltage-regulator fault that ended the mission after forty three days (Johnson, Morgan, and Rocklin 1967). For anyone trying to navigate this body of work, the annotated bibliography compiled in 1966 remains the single best index to the hundreds of reports the program generated (Atomics International 1966).

What this literature establishes is a regime that was rigorous in its engineering but improvised in its governance. There was no statute or directive that set quantitative pass-fail criteria. Authority flowed from the Atomic Energy Act of 1954, and review was handled by a panel assembled for the occasion. The safety standard that does appear in the technical work is deterministic and population based. The clearest statement of it is the reentry-burnup analysis that adopted the International Commission on Radiological Protection limit of two rem over thirty years and recommended that SNAP applications be held to a quarter of a rem over thirty years (Willis 1965). The reentry and disposal studies that sit alongside it, including the aerospace safety reentry program, treated the question as one of bounding the dose if the reactor came down, not of demonstrating a controlled disposal system (Elliott 1963). The AEC's own evaluation of the SNAP program rounds out the period as a statement of agency thinking rather than a binding rule (U.S. Atomic Energy Commission 1964).

The codification of the federal framework

The transition from this improvised regime to a codified one is best documented not in the primary sources but in a small secondary literature. The most useful single account is the Institute for Defense Analyses study by Behrens and colleagues, which states plainly that a high-level launch approval procedure was first formalized only in 1977 with Presidential Directive NSC-25, and which is candid that the modern process inherited from radioisotope missions may not transfer cleanly to fission reactors (Behrens et al. 2018). This is one of the few pieces of analysis that treats the history as a problem to be learned from rather than a story to be retold. The National Environmental Policy Act of 1969 added an environmental review layer that simply did not exist for SNAP-10A, and the layering of NEPA, PD/NSC-25, NSPM-20, and Space Policy Directive-6 is now the backbone of the system (U.S. White House 2019; U.S. White House 2020).

The contemporary NSPM-20 literature

The modern literature is dominated by the documents that implement NSPM-20 and by agency descriptions of them. The most cited analytical piece is the Sandia-authored article on implementing the memorandum for civil missions, which works through the tier structure and the risk-informed criteria and is effectively the reference text for how the new metrics are meant to function (Voss et al. 2021). On the regulatory side, the Nuclear Regulatory Commission's literature review is the most comprehensive survey of the approval process and is itself a useful map of the field (U.S. Nuclear Regulatory Commission 2023). NASA's procedural requirement is the operative implementation for civil missions and is where the abstract policy becomes concrete obligation, including the tiered safety analysis, the role of the standing review board, and the disposal clause that requires a highly reliable operational system for controlled disposition of a reactor in low Earth orbit (NASA 2022). NASA's own account of how it has reworked its safety program to infuse risk-informed thinking rounds out the picture (NASA 2023). What is striking about this body of work is how much of it is produced by the agencies that the rules govern. It explains the system clearly, but it rarely steps back to question whether the system is well designed.

Cross-cutting themes and debates

A few themes run through both bodies of literature and deserve to be drawn out. The first is the shift in how safety is even measured. The 1960s work fixed a population dose ceiling and asked whether the expected dose stayed under it (Willis 1965), whereas the modern regime asks a probabilistic question, namely whether there is a one in a million chance or greater of exposing any single member of the public to more than twenty five rem (U.S. White House 2019). This is a genuine change in the philosophy of safety, and it is underexamined. The implementation literature describes the new metric but does not really interrogate what is lost or gained by moving from a deterministic population standard to a probabilistic individual one.

The second theme is fuel. SNAP-10A ran on highly enriched uranium, which the period documents treated mainly as a criticality and material-at-risk concern (Atomics International 1965). The modern literature folds enrichment into tiering and into a broader security and nonproliferation conversation, and the policy drift toward low-enriched fuel is visible but not deeply theorized in the launch-safety sources specifically. The third theme is disposal, and here the literature is genuinely thin. The modern rule requires a sufficiently high disposal orbit together with a reliable system for controlled disposition of the reactor (NASA 2022). SNAP-10A is an instructive case precisely because its operational orbit was also its disposal orbit, reached by the launch vehicle and verified before the reactor was started, so the reactor ended its life shut down and nuclearly safe in a stable, long-lived orbit. The fault that ended the mission was in the spacecraft payload rather than the reactor, and it left the nuclear safety and the disposal intact. Whether that arrangement satisfies the modern reliability language is therefore a live question rather than a settled failure, and almost no published work runs it through. The wrinkle the literature does underplay is orbital debris, since the spacecraft later released NaK coolant and shed fragments, which is exactly the kind of outcome the current rule's orbital debris provisions would scrutinize even though the radiological disposal held. The fourth theme is the commercial pathway, which has no historical analog at all and which the existing literature barely addresses beyond noting that it now exists.

Gaps in the literature

Three gaps stand out. The first is the shortage of independent critique. The strongest analytical voices, the IDA study and the Sandia implementation article, are valuable precisely because they are uncommon, and most of the rest of the corpus is agency self-description (Behrens et al. 2018; Voss et al. 2021). A policy analyst entering this field will find plenty of explanation and very little adversarial assessment. The second gap is that disposal reliability, which is the requirement most likely to constrain a real reactor mission, is asserted more than it is studied. There is no substantial body of work analyzing what counts as a sufficiently reliable disposal system or how a designer would demonstrate it. The third gap is the disconnection between the historical and contemporary literatures. SNAP-10A is constantly cited as heritage, but it is rarely run through the current rules as a worked example, even though it is the only real data point the field has.

Conclusion

Read together, these two bodies of literature tell a coherent story even though their authors were not in conversation. The engineering rigor was present from the start, and the operating concept of launching cold and starting in a safe orbit has survived intact. What changed is the governance around it, from an improvised, deterministic, population-based regime to a codified, probabilistic, tiered one that now reaches into operation and disposal. The most useful contribution an analyst could make is not another description of NSPM-20 but the work the existing literature keeps gesturing at and never completes, namely a rigorous treatment of disposal reliability and a clear-eyed application of the modern standard to the one reactor the country has actually flown.

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